



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2007
Year 11
Half Yearly Exam

Mathematics Extension 1 - Continuers

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- All necessary working should be shown in every question.
- Hand in your answers in 2 separate bundles. Section A and Section B as indicated in the paper.

Total Marks – 80

- Attempt questions 1-6
- All questions are **NOT** of equal value.

Examiner: *F. Nesbitt*

SECTION A

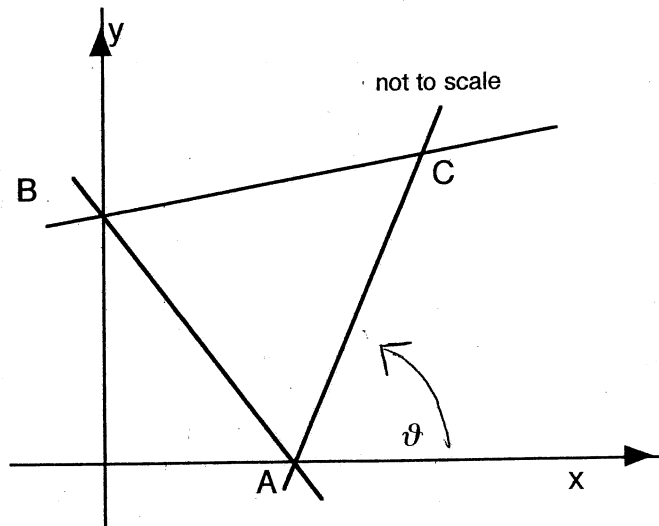
Question 1 (13 marks)

- (a) Simplify $3 - 4(x - 5)$ 1
- (b) Write 7 546 423 in scientific notation correct to 3 significant figures. 1
- (c) (i) Show that the function $f(x) = 2x^6 - 3x^4 + x^2$ is an even function. 2
- (ii) Find $f(\sqrt{x})$ 2
- (d) Solve $|x - 2| = 3$ 2
- (e) Find the value of $\frac{3.24}{\sqrt{5.13 - 1.89}}$ 1
- Give your answer correct to 2 decimal places.
- (f) Find the exact value of
- (i) $\sin 600^\circ$ (ii) $\tan \frac{2\pi}{3}$ 2
- (g) A triangle has sides 4.8 cm, 6.6 cm and 3.2 cm. Use the cosine rule to find the size of the largest angle to the nearest minute. 2

Question 2 (10 marks)

- (a) For the function $f(x) = x^2 - 5x + 6$
- (i) Find its x and y intercepts. 2
- (ii) Sketch the function. 1
- (iii) Find its Domain and range. 2
- (b) (i) Solve $\frac{2}{x} < 3$ 2
- (ii) Graph the solution on the number line. 1
- (c) Find the exact value of $\sin \frac{2\pi}{3} - \cos \frac{\pi}{4}$. Give your answer as a single fraction. 2

Question 3 (14 marks)



The points A, B, and C have coordinates $(4, 0)$, $(0, 5)$ and $(5, 9)$. The angle between AC and the x axis is θ .

- | | | |
|-----|---|---|
| (a) | Copy this diagram into your answer booklet. | 1 |
| (b) | Find the gradient of the line AC. | 1 |
| (c) | Find the size of the angle θ to the nearest degree. | 1 |
| (d) | Find, in general form, the equation of the line AC. | 2 |
| (e) | Find the coordinates of D, the midpoint of AC. | 1 |
| (f) | Show that BD is perpendicular to AC and state what this shows about $\triangle ABC$. | 2 |
| (g) | Find the area of $\triangle ABC$. | 2 |
| (h) | Find a point E on BD extended such that $BD = DE$. | 2 |
| (i) | What is the shape of the quadrilateral ABCE? Give a reason. | 2 |

SECTION B (Start a new booklet)

Question 4 (14 marks)

4. (a) Write with a rational denominator $\frac{1}{\sqrt{5}-2}$ 2
- (b) Simplify fully $\frac{m^2 - m}{m^2 - 1}$ 2
- (c) Write as a single simplified fraction $\frac{1}{1-x^2} + \frac{1}{1+x}$ 2
- (d) Write $0.\dot{3}\dot{1}$ in the form $\frac{m}{n}$ where m and n are integers. 1
- (e) Each interior angle in a regular polygon is 135° . How many sides does the polygon have? 2
- (f) A function is defined as:
 $f(x) = 6$ for $x < 3$
 $f(x) = 2x$ for $x \geq 3$
 Find (i) $f(4)$ 1
 (ii) $f(0)$ 1
 (iii) $f(-2)$ 1
- (g) If the point $(2, k)$ lies on the line $2y - 3x + 5 = 0$
 find the value of k . 2

Question 5 (15 marks)

5. (a) By completing the square, find the centre and radius of the circle

$$x^2 + y^2 - 4x + 10y - 7 = 0$$

3

- (b) An interval PQ has gradient - 3. A second interval passes through A (-2, 4) and B (1, k). Find the value of k if AB is parallel to PQ.

2

- (c) Factorise completely

(i) $u^2w + vw - u^2x - vx$

2

(ii) $27 - x^3$

1

- (d) Expand and simplify $2\sqrt{2}(\sqrt{5} - \sqrt{2})$

1

- (e) Write $\sqrt{500}$ in the form $a\sqrt{b}$

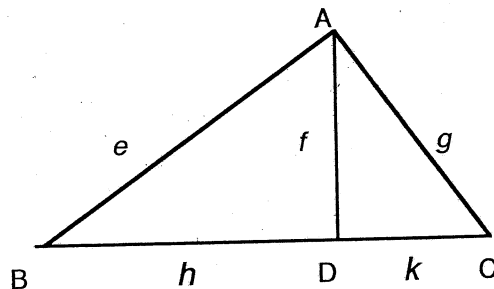
1

- (f) Find the point of intersection of the lines

$$x + y = 2 \quad \text{and} \quad 4x - y = 13$$

2

- (g)



In the triangle ABC above, the angle BAC is a right angle. AD is perpendicular to BC. Use Pythagoras' theorem to show that

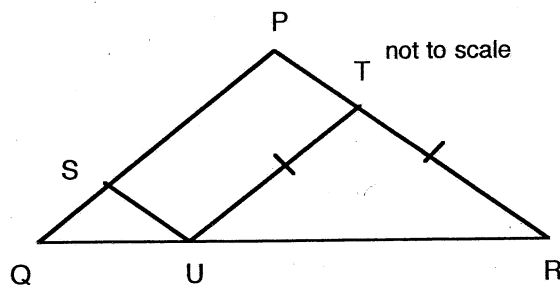
$$f^2 = hk$$

3

QUESTION 6**(14 marks)**

6. (a) Find the equation of the locus of the point $P(x, y)$ so that it is always the same distance from the point $(2, 1)$ as from the point $(1, 5)$ 2
- (b) Find the equation of a line passing through the intersection of the lines $x + 2y - 6 = 0$ and $3x - 2y - 6 = 0$ and passing through the point $(2, -1)$ 3
- (c) (i) On the same set of axes, draw graphs of the functions $y = |2x - 5|$ and $y = x + 2$ 2
- (ii) Use the graphs to solve the inequality $|2x - 5| \geq x + 2$ 2

(d)



In the triangle PQR above, $TU = TR$, SP is parallel to UT and

PT is parallel to SU .

- (i) Prove that the triangle PQR is isosceles 3
- (ii) If angle QPR is three times the size of the angle PRQ, prove that the angle PRQ = 36° 2

END OF PAPER

Section A.

Question 1:

$$\begin{aligned} \text{a) } 3-4(x-5) \\ = 3-4x+20 \\ = 23-4x \text{ (1)} \end{aligned}$$

$$\begin{aligned} \text{cii) } f(\sqrt{x}) &= 2(\sqrt{x})^6 - 3(\sqrt{x})^4 + (\sqrt{x})^2 \\ &= 2(x^{1/2})^6 - 3(x^{1/2})^4 + (x^{1/2})^2 \\ &= 2x^3 - 3x^2 + x \text{ (2)} \end{aligned}$$

$$\text{b) } 7.55 \times 10^6 \text{ (1)}$$

$$\text{c) i) } f(x) = 2x^6 - 3x^4 + x^2$$

$$\begin{aligned} f(-x) &= 2(-x)^6 - 3(-x)^4 + (-x)^2 \\ &= 2x^6 - 3x^4 + x^2 \end{aligned}$$

$$\therefore f(x) = f(-x) \therefore \text{even function (2)}$$

$$\text{d) } |x-2| = 3$$

the case

$$x-2 = 3$$

$$x = 5 \text{ (1)}$$

-ve case

$$-x+2 = 3$$

$$-x = 1$$

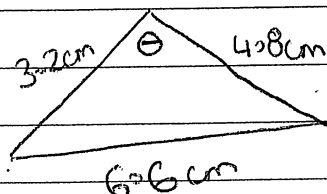
$$x = -1 \text{ (1)}$$

$$\text{e) } \frac{3 \cdot 24}{\sqrt{5 \cdot 3 - 1 \cdot 89}} = \frac{3 \cdot 24}{\sqrt{3 \cdot 24}} = \frac{3 \cdot 24}{1.8} = 1.80 \text{ (2dp). (1)}$$

$$\text{f) i) } \sin 600^\circ = -\frac{\sqrt{3}}{2} \text{ (1)}$$

$$\text{ii) } \tan \frac{2\pi}{3} = -\frac{\sqrt{3}}{3} \text{ (1)}$$

$$\text{g) } 4.8 \text{ cm, } 6.6 \text{ cm, } 3.2 \text{ cm}$$



(not to scale)

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\begin{aligned} \cos A &= \frac{(4.8)^2 + (3.2)^2 - (6.6)^2}{2 \times 4.8 \times 3.2} \\ &= -0.334635416 \end{aligned}$$

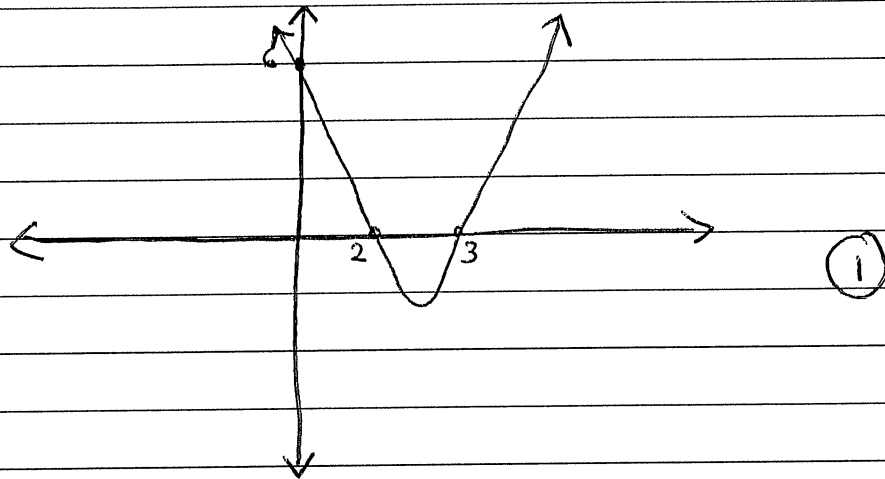
$$\therefore A = 109^\circ 33' \text{ (2)}$$

Question 2:

a) i) x -intercepts: $0 = x^2 - 5x + 6$
 $0 = (x-3)(x-2)$
 $\therefore x = 3, x = 2$ (1)

y -intercepts: $y = 0 - 0 + 6$
 $y = 6$ (1)

ii)



iii) Domain: All real x . (1)

Range: $y \geq -1/4$ (1) axis of symmetry is half way between 2 + 3 $\therefore 2.5$

$$f(2.5) = 2.5^2 - 5 \times 2.5 + 6$$

$$= -1/4$$

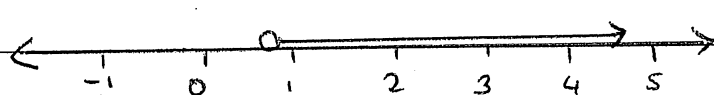
b) i) solve $\frac{2}{x} < 3$

$$2 < 3x$$

$$1 \text{ for } x < 2/3$$

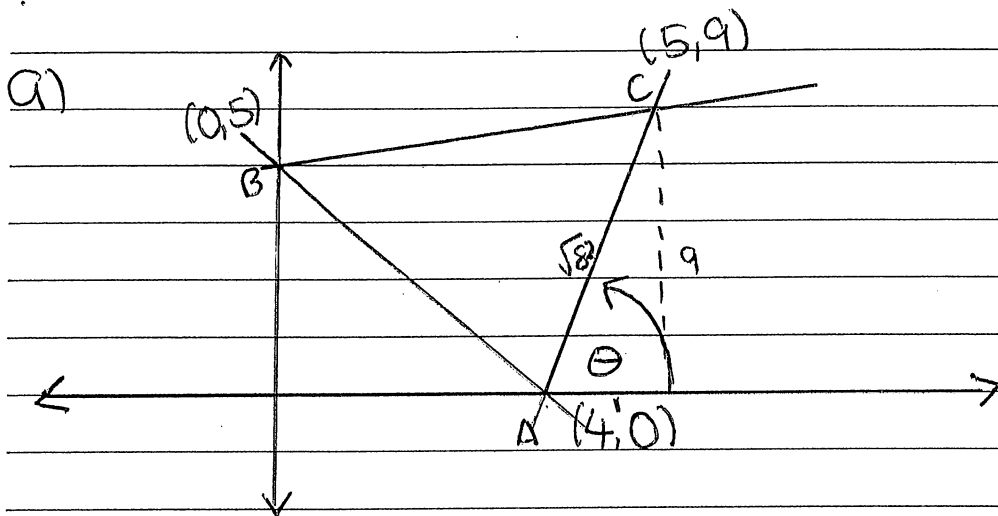
$$x > 2/3 \quad (2)$$

ii)



c) $\sin \frac{2\pi}{3} - \cos \frac{\pi}{4} = \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} = \frac{\sqrt{6}-2}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$
 $\frac{\sqrt{2}-2\sqrt{2}}{4} = \frac{\sqrt{3}-2\sqrt{2}}{4} = \frac{\sqrt{2}}{2}$

Question 3:



b) $m_{AC} = \frac{9-0}{5-4}$

$$= \frac{9}{1}$$

$$= 9$$

①

c) Distance of AC = $\sqrt{(5-4)^2 + (9-0)^2}$

$$= \sqrt{1 + 81}$$

$$= \sqrt{82}$$

$$\cos \theta = \frac{1^2 + \sqrt{82}^2 - 9^2}{2 \times 1 \times \sqrt{82}}$$

$$= 0.110431526$$

$$\therefore \theta = 83.65980825$$

$$= 84^\circ \text{ (nearest degree)}$$

①

d) AC: $y-9 = 9(x-5)$

$$y-9 = 9x-45$$

$$9x-y-36=0$$

②

e) $D = \left(\frac{5+4}{2}, \frac{9+0}{2} \right)$

$$= \left(\frac{9}{2}, \frac{9}{2} \right)$$

①

$$f) m_{BD} = \frac{9/2 - 5}{9/2 - 0} = -1/9$$

$$m_1 \times m_2 = 9 \times -1/9 = -1$$

$\therefore BD \perp AC$ (1)

$\therefore \triangle ABC$ is isosceles as $\triangle ABD \equiv \triangle BCD$
 $\therefore BC = BA$. (1)

$$g) \text{ Distance of } BD = \sqrt{(0 - 9/2)^2 + (5 - 9/2)^2} \\ = \sqrt{81/4 + 1/4} \\ = \sqrt{82/4} \\ = \sqrt{82}/2$$

$$\therefore \text{Area of } \triangle ABC = \frac{1}{2} \times \sqrt{82} \times \frac{\sqrt{82}}{2} \\ = \frac{82}{4} \\ = 20\frac{1}{2} \text{ u}^2 \quad (2)$$

h) Find E such that $BD = DE$
 $\therefore D$ is the midpoint of BE .

$$\therefore \left(\frac{9}{2}, \frac{9}{2}\right) = \left(\frac{0+x}{2}, \frac{5+y}{2}\right)$$

$$\therefore x = 9 \quad \begin{aligned} 5+y &= 9 \\ y &= 4 \end{aligned}$$

$\therefore$ the coordinates of E are (9, 4) (2)

i) $ABCE$ is a rhombus.

$$\triangle ABC \equiv \triangle ACE$$

$$\therefore BC = BA = EC = EA \quad (2)$$

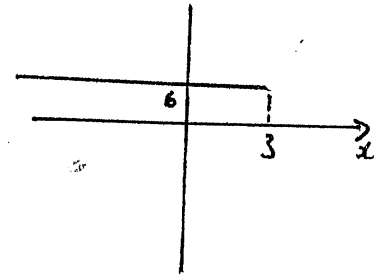
$\therefore$ diagonals bisect at $\perp$.

4a) $\frac{1}{\sqrt{5}-2} = \frac{1}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2}$ (1 for rationalising f) $f(x) = 6 \quad x < 3$
 $= \frac{\sqrt{5}+2}{1}$ $f(x) = 2x \quad x \geq 3$

$$= \sqrt{5}+2 \quad (2)$$

b) $\frac{m^2-m}{m^2-1} = \frac{m(m-1)}{(m-1)(m+1)}$ $\frac{1}{2}$ each for factors (both)

$$= \frac{m}{m+1} \quad (2)$$



$$f(4) = 8 \quad (1)$$

$$f(0) = 6 \quad (1)$$

$$f(-2) = 6 \quad (1)$$

c) $\frac{1}{1-x^2} + \frac{1}{1+x} = \frac{1}{(1-x)(1+x)} + \frac{1}{1+x}$

$$= \frac{1 + 1-x}{(1-x)(1+x)}$$

$$= \frac{2-x}{(1-x)(1+x)} \quad (2)$$

g) $2y - 3x + 5 = 0$
 Since $(2, 2)$ lies on line

$$2k - 6 + 5 = 0$$

$$2k = 1$$

$$k = \frac{1}{2} \quad (2)$$

Must show working

d) $x = 0.313131 \dots$

$$100x = 31.3131 \dots$$

$$99x = 31$$

$$x = \frac{31}{99} \quad (1)$$

e) $\frac{180(n-2)}{n} = 135$

$$180(n-2) = 135n$$

$$180n - 360 = 135n$$

$$45n = 360$$

$$n = 8 \quad (2)$$

QUESTION FIVE (15)

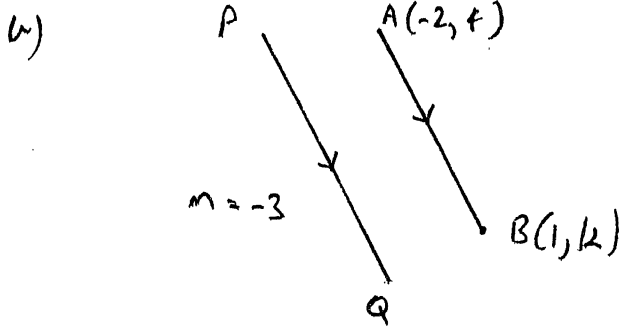
a) $x^2 + y^2 - 4x + 10y - 7 = 0$

$$x^2 - 4x + y^2 + 10y = 7$$

$$x^2 - 4x + 4 + y^2 + 10y + 25 = 36 \quad (1)$$

$$(x-2)^2 + (y+5)^2 = 36 \quad (1)$$

Circle centre $(2, -5)$ radius 6 (3)



Eqn of AB

$$y - 4 = -3(x + 2)$$

$$y - 4 = -3x - 6$$

$$3x + y + 2 = 0$$

Since $(1, k)$ lies on this line

$$3 + k + 2 = 0$$

$$k = -5 \quad (2)$$

c) $u^2w + vw - u^2x - vx$

$$= w(u^2 + v) - x(u^2 + v)$$

$$= (w - x)(u^2 + v) \quad (2)$$

iii) $27 - x^3 = 3^3 - x^3$

$$= (3 - x)(9 + 3x + x^2) \quad (1)$$

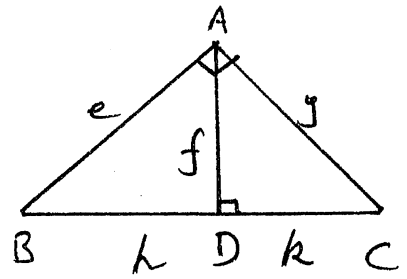
d) $2\sqrt{2}(\sqrt{5} - \sqrt{2}) = 2\sqrt{10} - 4 \quad (1)$

e) $\sqrt{500} = \sqrt{100} \times \sqrt{5}$
 $= 10\sqrt{5} \quad (1)$

f)

$$\begin{aligned} x + y &= 2 \\ 4x - y &= 13 \\ 5x &= 15 \\ x &= 3 \\ y &= -1 \end{aligned} \quad (2)$$

g)



$$e^2 + g^2 = (h + k)^2$$

$$e^2 + g^2 = h^2 + 2hk + k^2 \quad (1)$$

Then

$$e^2 = f^2 + h^2$$

$$g^2 = f^2 + k^2$$

In (1)

$$f^2 + h^2 + f^2 + k^2 = h^2 + 2hk + k^2$$

$$2f^2 = 2hk$$

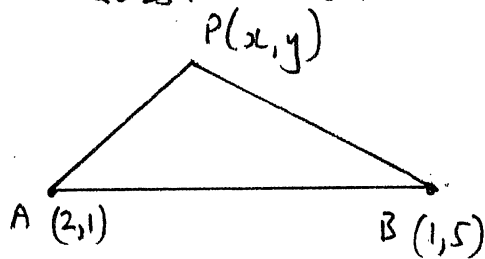
$$f^2 = hk \quad (3)$$

h) $\frac{4 - k}{-2 - 1} = -3$

$$\begin{aligned} 4 - k &= 9 \\ k &= -5 \end{aligned}$$

QUESTION SIX.

14



$$PA = PB$$

$$\sqrt{(x-2)^2 + (y-1)^2} = \sqrt{(x-1)^2 + (y-5)^2}$$

$$x^2 - 4x + 4 + y^2 - 2y + 1 = x^2 - 2x + 1 + y^2 - 10y + 25$$

$$-4x + 2y + 5 = -2x - 10y + 26$$

$$-2x + 8y - 21 = 0$$

$$2x - 8y + 21 = 0 \quad 2$$

OR Locus of P is the perp.

bisector of AB

Mid point of AB $(\frac{3}{2}, 3)$

Gradient of AB -4

Locus of P

$$y - 3 = +\frac{1}{4}(x - \frac{3}{2})$$

$$4y - 12 = x - \frac{3}{2}$$

$$8y - 24 = 2x - 3$$

$$2x - 8y + 21 = 0$$

b) Required line has eqn.

$$x + 2y - 6 + k(3x - 2y - 6) = 0$$

Since $(2,1)$ lies on line.

$$2 - 2 - 6 + k(6 + 2 - 6) = 0$$

$$2k = 6$$

$$k = 3$$

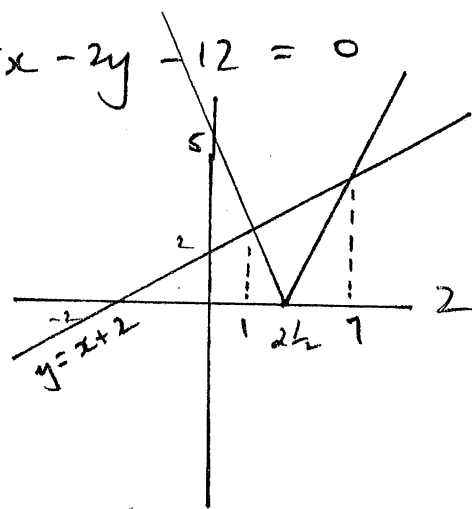
Required line is

$$x + 2y - 6 + 3(3x - 2y - 6) = 0$$

$$x + 2y - 6 + 9x - 6y - 18 = 0$$

$$5x - 2y - 12 = 0$$

c)



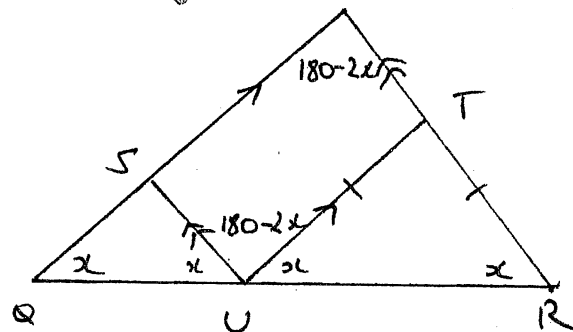
$$x \geq 7$$

$$\text{and } x \leq 1 \quad 2$$

Need to use graphs

(only if graphs not used.)

d)



$$\text{Let } \angle PQR = x$$

$$\text{Then } \angle SQR = x \text{ (corresponding)}$$

$$\angle TUR = x \text{ (also } \Delta)$$

$$\angle SUT = 180 - 2x$$

$$\angle QPR = 180 - 2x \text{ (opp angles of } \Delta \text{ PQR)}$$

$$\angle PQR = x$$

Angle Sum of ΔPQR

Given

$$\angle QPR = 3 \times \angle PQR$$

$$180 - 2x = 3x$$

$$5x = 180$$

$$x = 36$$

2